Attributing Construction Delay

A problem statement for the construction root-cause project

two independent verified searches · prep for the Adrian meeting · 2026-09-10

The project survives the "this is already solved" objection — but not by abandoning root cause analysis. It survives by scaling and measuring it. This note sets out what RCA genuinely provides, what it structurally cannot, what we would add, what data that requires, what of it exists, and how we would know if it worked.

Status: draft for discussion. **Stance:** assume the project proceeds. **Open question:** differentiation.

1. The objection, stated fairly

Root cause analysis is not an open problem. There are at least six books on it, a certification course runs about $400 in Canada, and the methodology — 5 Whys, Ishikawa, FMEA, fault tree, Pareto — has been stable for decades. At least three enterprise construction platforms already ship RCA modules. The chief scientist put the differentiation question to the head of construction and got no answer.

The challenge is legitimate and it will be asked again by someone more senior. It also has a sharper edge than it first appears, and any answer that misstates what these methods do will be punctured in the first five minutes.

> **Correction worth making before the meeting.** It is tempting to say classical RCA "produces no quantity and no counterfactual." That is true of 5 Whys and Ishikawa. It is **false** for the other three, and a knowledgeable reviewer will say so.
>
> **Fault tree analysis is quantitative by construction:** basic-event probabilities roll up to a top-event probability, and its importance measures — Birnbaum, Fussell-Vesely — express precisely the fraction of top-event probability attributable to a given basic event. Its **minimal cut sets are an enumerated set of counterfactual interventions. FMEA** produces a Risk Priority Number from severity × occurrence × detection. **Pareto** is a portfolio base rate by definition.

So the differentiator is narrower and must be stated precisely. Those methods quantify only for **engineered systems with known structure and elicited, approximately independent basic-event probabilities**. They carry no theory of *observational* estimation — no treatment of confounding, of selection, or of inference from data that was collected rather than designed. A construction portfolio is an observational setting with an unknown graph. That is the gap, and it is a gap in inference, not in method.

2. What root cause analysis actually gives us

RCA is not the thing being replaced. It is the instrument the system is built around, and it contributes four things we cannot obtain elsewhere.

The hypothesis space. A causal model needs candidate causes to reason over. Ishikawa's categories and the DOT reason codes in §7 are not a competitor to the model — they are its label

schema, developed over decades and already defended in practice. Inventing our own would be strictly worse. Ishikawa's job here is a *prior over hypotheses*, not inference.

The graph structure. In Pearl's framework, causal *identification* is a property of the graph; only *estimation* is statistical. With a few hundred projects, roughly thirty causes, heavy missingness and selection on the outcome, we cannot learn a DAG from data — causal discovery at that scale in a non-stationary domain returns confident nonsense. The graph must be elicited from domain experts, and the elicitation instruments for a causal skeleton are exactly Ishikawa, fault-tree structure, and control-structure analysis. **RCA supplies the structure our model estimates over.**

The labelling protocol. §13 establishes that no corpus of adjudicated causal labels exists. Something has to manufacture them, and what manufactures a defensible causal label is a disciplined human method: 5 Whys to push past the proximate cause, fault-tree logic to test whether the chain is complete, AACE 29R-03 to adjudicate. Option F depends on this entirely.

The evaluation rubric. A well-formed root-cause statement has testable properties: it passes a necessity test, it carries an evidence reference, it terminates at a *controllable* cause, and it pairs a **condition** (weather, a latent site state) with an **action** (the decision to proceed without float). These are scoreable, which is the direct fix for §8's "cannot be scored" problem.

Each method also contributes something specific that a statistical attribution model does not:

Method	What it uniquely provides
5 Whys	Terminates at a *specific, actionable* cause — "the PO went out 11 days late because the submittal sat with the architect" — not a taxonomy node. Answers §12 Q1 in a way a coefficient cannot. Open vocabulary: it can surface a cause absent from the taxonomy.
Ishikawa	Forces coverage of the hypothesis space *before* evidence narrows it. The antidote to availability bias.
Fault tree	Boolean structure with AND gates — the correct representation of **concurrent delay** (both had to happen) versus alternative causes (either sufficed). This is the algebra for the concurrency allocation that SCL and AACE practitioners argue about.
FMEA	Prospective, not retrospective. Its Detection column asks how we would know a failure mode was developing — the specification for §11's leading indicators.
Pareto	Tells us which causes deserve the labelling budget. Given labels are the bottleneck, this decides where adjudication effort goes.

What RCA structurally cannot do

It never sees the denominator. A 5 Whys session convenes only over incidents. It therefore cannot compute P(cause present | nothing slipped), and no amount of facilitator skill fixes this — the base-rate blindness is built into the method's sampling frame. Only a population dataset addresses it.

Three further limits are documented rather than asserted:

- **Low reproducibility.** Different teams analysing the same incident reach different causes, and resulting action plans skew toward weak interventions — "retrain," "remind." (Peerally et

al., *BMJ Qual Saf* 2017.) Our inter-rater experiment in §11 is a measurement *of RCA quality*, not a competitor to RCA.

- **Stopping-rule bias.** A 5 Whys chain terminates where the facilitator's authority or knowledge runs out, not where causation does. (Card, *BMJ Qual Saf* 2017.)
- **Reporting selection.** Only incidents someone wrote up are ever analysed. The activities that quietly slipped never enter the sample.

The one-sentence answer to "how does RCA help?" It supplies the taxonomy, the graph structure, the adjudication protocol that generates our labels, and the rubric that scores our model. What it cannot supply — and what we add — is the denominator, statistical separation of concurrent causes, and an answer that arrives in week one rather than month nine.

3. The reframe

"Use AI to do root cause analysis" is the wrong slogan, not because RCA is inadequate but because it describes the instrument rather than the contribution. The defensible object:

Problem statement. Given a project's baseline schedule, its as-built record, and the multimodal evidence stream generated on site, **attribute realized schedule and cost variance at activity level to a labelled set of causal factors** — with a magnitude and an uncertainty per factor, an explicit causal structure separating initiating causes from propagated consequences, and enough lead time that the attribution can still change a decision.

It works if it changes what a project manager does, and survives challenge from a party with money at stake.

Stated against RCA rather than instead of it, what we add is:

- **A denominator.** Base rates across a portfolio, which is what bidding, contingency and pricing actually consume — and which no incident-convened method can produce.
- **Observational estimation.** Separating initiating cause from propagated consequence in data that was collected, not designed, with confounding and selection handled explicitly.
- **Throughput.** A disciplined RCA on every activity that slips, in days rather than months — not once per incident, months late, when a facilitator is available.
- **Magnitude with uncertainty.** "Supply delay accounts for 14 ± 4 of the 31 days lost, of which 9 sat on the critical path."

That framing survives the "already solved" challenge better than a teardown does, because it requires no claim that the incumbents got it wrong — only that they cannot run the method often enough, or compute the denominator at all.

4. Why naive attribution gets the wrong answer

This is the technical heart of the case, and the argument that makes "path-breaking" defensible rather than hopeful.

As Figure 1 shows, a correlational model sees rework as the dominant correlate of overrun and reports it as the root cause. This is precisely the failure 5 Whys exists to prevent in humans, and precisely the failure a naive classifier reintroduces at scale and with false confidence.

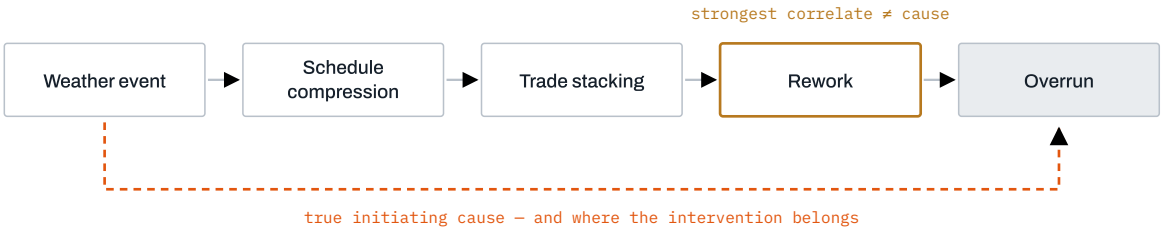


Figure 1. The confounding chain. A weather event drives schedule compression, which drives trade stacking, which drives rework, which drives overrun. Rework is the strongest correlate of overrun, so a correlational model reports it as the root cause — and the intervention that follows, tightening workmanship QA, is the wrong one. The lever was float protection and resequencing after the weather event.

Two caveats belong with the figure, and stating them ourselves is stronger than having them raised:

Collider bias. Our unit of analysis (§6) is defined by *deviation beyond a threshold*. Training only on events selected on the outcome induces spurious negative correlation between causes — Berkson's paradox. A human adjudicating a single event estimates nothing and cannot fall into this; our portfolio model can, by construction. Any estimator must handle selection explicitly, and §11's trigger-based random sampling is the practical mitigation.

The figure is a chain-of-events model. Linear cause-chains are exactly what systems-safety work (Leveson's STAMP, and CAST for incident analysis) argues against, on the grounds that most losses arise from *inadequate control* rather than a chain of component failures. Modelled as a control structure — owner → GC → sub → crew, with the designer–RFI feedback loop — the transferable flaw categories are recognisable immediately: missing feedback (progress statused to plan), inadequate control algorithm (float consumed silently), process-model mismatch (the PM believes the steel is on order). That yields a better graph skeleton than a fishbone, and its prospective form (STPA) specifies §11's leading indicators as *signals that a controller's process model has diverged from the plant*.

5. Two tracks, not one

The team said delays **and defects**. These are different inferential problems and collapsing them is a mistake this document previously made.

	Delay attribution	Defect diagnosis
Question	Which causes account for how many days, across a portfolio	Why did *this* wetspot occur — membrane, plumbing, or rain during the open-roof period
Nature	Statistical, population-level	Forensic, mechanistic, single-case
Right method	Causal estimation over an elicited graph	Differential diagnosis plus a discriminating test
Evidence	Schedules, logs, exogenous covariates	Site inspection, water test, thermography, as-built detail
Correct protocol	Attribution with uncertainty	**Hypothesis list, then a test that discriminates**

The second column matters for the meeting: **hypothesis-list-then-test is the correct protocol for defect causation.** Nobody attributes a membrane failure with a regression. The team's instinct is right here — the deficiency is that the loop stops at the hypothesis list and never reaches the discriminating test, and never records the outcome.

6. Unit of analysis

Most projects of this kind fail quietly here, before any modelling starts. The unit must be fixed first, because it determines what can be counted.

The unit is the **activity-level variance event**: a WBS activity whose actual dates or duration deviate from baseline beyond a threshold, carrying its critical-path status at the moment of deviation.

- **Not the project.** Too coarse — causes superimpose, and sample size collapses to the number of projects.
- **Not the reported incident.** Biased to whatever someone bothered to write up.

Note that defining the unit by deviation is what creates the collider problem in §4. It is the right unit and it carries that cost; the cost must be managed, not ignored.

7. Data specification

What §3 requires, layer by layer. The last row decides whether this is feasible.

Layer	Fields	Source	Availability
Baseline	Activity ID, WBS, planned dates, duration, predecessors and lags, total float, resource loading	P6 / MS Project export	Exists; rarely shared
As-built	Actual start and finish, progress curve, baseline revision history	Same	Exists; revisions often lost
Variance	Δduration, Δdates, critical-path status at deviation, downstream propagation	Derived	Must be computed
Evidence — structured	RFI open/close latency, change orders with reason codes, submittals, PO vs. receipt dates, inspections, labour hours by trade	PM system	Exists; siloed
Evidence — unstructured	Daily logs, timestamped site photos and video, meeting minutes, punch lists	Field apps	High volume, no structure
Exogenous	Station-level hourly weather, permit and inspection turnaround, material price and lead-time indices	Public	**Free — the cheap layer**
Labels	Root cause per event, days attributed per factor, labeller identity, confidence, contested flag	Expert adjudication	**Scarce — the bottleneck**

> **The binding constraint.** Everything above the last row is an engineering problem. The last row is the project. Labelled, magnitude-attributed root causes at activity level barely exist in public, and the firms holding the best data have the least incentive to share — the Applied Materials refusal generalizes.

Cause taxonomy

Fixed before labelling begins; changing it mid-project invalidates every label collected to that point. Version the taxonomy and stamp every label with the version.

Group	Causes
Design	Incomplete or incorrect documents, RFI latency, late design change
Procurement	Lead-time miss, vendor default, logistics failure
Labour	Shortage, productivity shortfall, absenteeism, skill mismatch
Equipment	Failure, availability, mobilisation delay
Site conditions	Differing subsurface, access restriction, adjacent works
Environment	Weather, seasonal constraint
Regulatory	Permit, inspection turnaround, code change
Coordination	Trade stacking, sequencing error, subcontractor interface
Quality	Rework from defect, failed inspection
Client	Scope change, decision latency, payment delay
Force majeure	Exceptional events as contractually defined

Distinguish three tiers within it, as forensic practice does: the **causal factor** (what happened here), the **root cause** (the correctable condition behind it), and the **generic cause** (the systemic pattern across projects). These map onto initiating / propagated / systemic and should be separate fields, not one.

One cause per activity row is the wrong shape

Forcing a single cause into each activity record breaks on the cases that matter most: one event affecting many activities, one activity with several causes, mitigation, and concurrent delay. Separate schedule state, disruption event, and causal attribution, and link them many-to-many.

- `activity_schedule_version` — project, schedule version, activity, WBS, baseline and current planned dates, actual dates, status date, calendar, total float, critical-path flag. Never overwrite a baseline or calendar.
- `delay_event` — event window, cause code, narrative, and the source document it came from. The cause must trace to evidence, never to a model's inference.
- `event_activity_impact` — the many-to-many link, with claimed, assessed and awarded days, critical-path effect, concurrency flag, allocation method. **This is a one-bit approximation of a fault-tree AND gate; use the FTA algebra where concurrency is contested.**
- `causal_adjudication` — cause code, causal days, decision-maker, evidence reference, and an explicit **attribution status**.

  The distinction that prevents a tautology. Critical-path contribution days are *not* causal days. "This activity contributed 9 days of critical-path delay" is a statement about the schedule network — arithmetic, and a tool that reports it has explained nothing. "Late steel delivery caused 9 days of this activity's slip" is a causal claim requiring evidence and an allocation rule for concurrency. Store the first as `schedule_contribution_days` and never let it enter a field named for causation.

  Label every attribution with how it was established — `observed`, `asserted`, `owner_accepted`, `expert_assessed`, `adjudicated` — and never treat those as

equivalent.

8. Where the space is actually open

Honest column by column. "Open" means no incumbent occupies it well, not that it is easy.

Capability	Codified RCA	Enterprise software	Photo → LLM	Verdict
Single-incident narrative	Yes	Yes	Yes	Crowded
Workflow, system of record	—	Yes	—	Owned
Quantified attribution, engineered systems	Yes (FTA/FMEA)	Partial	—	Solved elsewhere
Quantified attribution, observational data	—	Partial	—	**Open**
Portfolio base rates / denominator	—	Reporting only	—	**Open**
Causal ID under confounding	Human judgment	—	—	**Open — strongest**
Prospective lead time	FMEA detection	Some vendors	—	Contested
Multimodal evidence fusion	—	—	Partial	**Open**
Contest-grade defensibility	Expert report	—	—	**Open — high burden**

The real problem with the current approach

Photograph the water damage, hand the video to GPT, get three or four hypotheses, let a human pick one. The damaging criticism is not that this is unoriginal — per §5, hypothesis-then-test is the *right* protocol for defect diagnosis. It is that **the loop is not instrumented**, so it cannot be scored and it accumulates nothing.

That is a logging problem, and it is fixable in about a week. Persist the full hypothesis set rather than the winner, an evidence link per hypothesis, the pick, labeller identity, confidence, a dissent flag, and — critically — **the discriminating test and its outcome**. Do that and the loop stops being a demo and becomes the label engine Option F requires.

> **Detection is the commodity; attribution is the work.** There is a second, quieter problem: the loop is a *worse detector* than what can be downloaded free. Real inspection datasets with water-damage labels exist (§13). A model fine-tuned on those identifies the defect more accurately and more cheaply than prose hypotheses, and returns a box or mask rather than a paragraph.
>
> *This is a wetspot* is available off the shelf. *This wetspot came from the failed membrane, not from rain during the open-roof period, and it cost nine days* is the entire problem — and no dataset in existence contains that sentence.

9. Scope options

	Option	What it is	Profile
A	Evidence structuring	Turn the unstructured site record into structured variance events with linked evidence.	Low ML risk, certain to ship, foundation
B	Retrospective quantified attribution	Portfolio base rates with magnitudes and uncertainty.	First result beyond the incumbents; needs schedule data at scale
C	Prospective early warning	Forecast the slip and the cause developing behind it.	Highest site value; crowded, data-hungry
D	Causal counterfactual engine	Causal graph over the schedule network.	Strongest USP; research risk
E	Claims-grade attribution	Force majeure versus negligence, to a contestable standard.	Highest value per unit; gate behind counsel
F	Build the corpus	Create the labelled corpus, taxonomy and evaluation protocol that does not exist.	Moat, publishable, incumbent-independent

Option D is cheaper than it looks. The counterfactual query set does not need inventing: **minimal cut sets of an elicited fault tree enumerate them.** "Had the steel landed on time, would the pour have slipped?" is a cut-set question, and the machinery is fifty years old.

Option F should be reshaped. The obstacle is not curation effort but data rights — CII refuses raw data to outsiders, nPlan holds its corpus, Applied Materials refused. Safety-critical domains solved exactly this problem with **confidential reporting through a trusted third party**: NASA's Aviation Safety Reporting System takes voluntary, confidential reports from pilots and controllers, with protection from civil penalty and certificate suspension, precisely because nobody will report to their own regulator or a competitor. Construction already has the pattern — **CROSS**, operated by Structural-Safety Ltd with IStructE, ICE and IFE across the UK, Australasia and the US, takes confidential built-environment reports and publishes them anonymised. CROSS covers structural and fire safety, *not* schedule delay, so this is a proven template rather than an existing corpus.

That reframes F from "curate public sources" to "**be the trusted third party for delay reporting**" — which answers the data-access risk in §12 structurally rather than by negotiation, and is far more defensible than another vendor asking firms for their schedules.

> **Recommendation. A → B, with F carried alongside, and D named as the research bet.** A makes the data legible; B produces the first number that beats an incumbent; F builds the asset nobody has; D is where the differentiation claim lands. Hold C until longitudinal data exists, and gate E behind counsel. Run the defect track (§5) in parallel — it delivers value on one site in weeks and needs nobody's permission.

10. Where each approach fails

Stating both directions is what makes the analysis credible.

Classical RCA failures that data fixes

- **No denominator.** The largest one. It cannot compute P(cause | no delay).

- **Stopping-rule bias.** Chains end at the limit of the facilitator's authority (Card 2017).
- **Hindsight and narrative bias.** The chosen cause is the one that makes a story.
- **Low reproducibility, weak actions.** Different teams, different causes; plans skew to "retrain, remind" (Peerally et al. 2017).
- **Reporting selection.** Only written-up incidents are analysed.

Causal-graph / ML failures that a disciplined 5 Whys avoids

- **Closed world.** A cause absent from the taxonomy is invisible. The sub went bankrupt because *their other client* did not pay — no node for that. A human names it in minute two.
- **Silent graph misspecification.** Omit the edge "the PM chose which activities to protect" and every estimate is confidently wrong with no internal signal.
- **Collider bias by construction.** See §4.
- **Positivity failure.** Force majeure has no counterfactual support in any dataset. Mechanism reasoning handles the one-off; the model cannot.
- **Measurement fiction.** P6 actuals are frequently statused to plan at month end. A model treats them as observations; a scheduler knows the update was cosmetic.
- **Goodhart, once it touches claims.** Daily logs get written for the dispute. Adjudicators know this; models do not.

11. How we would know it worked

The measurement nobody has taken

Have three experienced PMs independently label the same fifty variance events. Their agreement rate is approximately the model's ceiling — if they agree 55% of the time, every accuracy claim above 55% is measurement error.

A week, no data acquisition, and it can run on data already downloadable (§13). It is the closest thing to a genuine feasibility gate, and it is also a measurement *of RCA reproducibility* — the same quantity Peerally et al. found wanting in healthcare.

Fix the sample, not just the model

Retrospective review of *reported* incidents inherits reporting selection. Healthcare's answer is the **Global Trigger Tool**: define triggers in the data, then adjudicate a **random sample** of triggered records rather than the ones someone escalated — an approach that found adverse events roughly ten times more common than voluntary reporting suggested (Classen et al., *Health Affairs* 2011).

The construction analogue is direct: define triggers in schedule data — float consumed beyond a threshold, actual-to-plan ratio, RFI ageing — and adjudicate a random sample of triggered activities. This is the cheapest available fix for survivorship bias and it should be in the first sprint.

Borrow the precursor framing

The NRC's **Accident Sequence Precursor** programme computes, for an observed operational event, the *conditional probability* that it leads to core damage — CCDP for initiating events, ΔCDP for degraded conditions — using fault-tree and event-tree models, with an event counting as a precursor above 10^{-6} and as significant above 10^{-3}. Forty years of practice at exactly our question, in a different domain.

Ours: **given this RFI-latency event, what was the conditional probability of critical-path slip?** That is a better-posed target than "predict the delay," it is estimable from far fewer labels, and it produces a naturally calibrated, thresholded severity scale.

Metrics

- **Attribution accuracy**, top-1 and top-3, always against the inter-rater ceiling rather than 100%.
- **Magnitude error** — MAE in days attributed per factor. The metric incumbents cannot report at all.
- **Calibration** — reliability curve and ECE. A confidently wrong answer is worse than an honestly uncertain one.
- **Conditional slip probability** given a precursor event, with lead time at fixed precision.
- **Action strength.** Score each attributed cause by the strength of the corrective action it enables. RCA² (National Patient Safety Foundation, 2015) provides an Action Hierarchy tool ranking actions by expected effectiveness; a cause that yields only "retrain the crew" is worth less than one yielding a scheduling constraint, and the metric should say so.

Baselines it must beat

1. The base rate — always predict the most common cause for that trade and phase.
2. The incumbent tool's output on the same events.
3. **An experienced PM reading the daily log for five minutes.**

If it cannot beat the third, there is no product. Worth saying aloud, because it is the baseline most likely to be skipped — and note that the team's own loop is a scalable approximation of exactly that PM.

12. Risks

- **Label scarcity.** The binding constraint is ground truth, not model capability. Most spend belongs in adjudication, not compute.
- **Confounding and collider bias.** See §4. Correlational attribution produces confident, wrong interventions.
- **Data access.** Applied Materials refused a hundred-billion-dollar company's data to an external party. Assume the same posture from every contractor with a good corpus — which is why §9's third-party reporting model matters.
- **Survivorship bias.** Mitigated by trigger-based random sampling (§11), not by more data.
- **Liability creep.** The moment output touches a claim, the standard shifts from "useful" to "defensible in a dispute."
- **Evaluation theatre.** Unscoreable outputs mean six months with nothing that could have been falsified.
- **Taxonomy drift.** Changing the label set mid-project invalidates everything collected before the change.
- **Closed-world blindness.** The model cannot report a cause the taxonomy lacks. Keep an open-vocabulary escape hatch in the adjudication form and mine it quarterly.

13. Questions to put to the team

In this order. The first three do most of the work.

1. What decision changes because of this output, and who makes it differently tomorrow?
2. How is that decision made today, and what does a wrong answer cost?
3. Which of the three incumbent tools are we differentiating against, and on which axis?
4. Is this the delay track, the defect track, or both — and do you agree they are different problems (§5)?
5. What is the unit of analysis — project, incident, or activity-level variance event?
6. What is the inter-rater agreement of our own experts on root cause? If unmeasured, can we measure it this month?
7. Is the output advisory, or will someone rely on it in a claim?
8. What data do we have rights to *today* — not what we hope to negotiate?
9. Are we logging the full hypothesis set, the evidence per hypothesis, and the discriminating test outcome? If not, why not, starting this week?
10. What does this do that a PM with the daily log cannot do in five minutes?

14. The dataset search — verified findings

Every source below was opened and read. Where something could not be verified it says so.

What exists and is usable

Source	What it gives you	Gap	Status
OR&S real project database, Ghent	231 empirical projects, construction dominant. Baseline network and resources, period tracking data, corrective actions on selected projects. Free.	No root-cause labels	Verified — the spine
AACE 29R-03 Forensic Schedule Analysis	The standardised adjudication practice: nine Method Implementation Protocols.	A method, not data	Verified — the labelling protocol
DOT change-order reason codes (Florida, Ohio)	Published coded cause taxonomies in production use.	Taxonomy, not events	Verified — validates §7
SODA	19,846 construction site images, 286,201 annotated objects, 15 classes.	No link to schedule	Verified — evidence layer
Weather, price and lead-time indices	Station-level historical weather; construction cost index.	Join keys need work	Free, straightforward

The headline result

> **Confirmed independently, twice.** There is **no public corpus of activity-level variance events carrying adjudicated root-cause labels with attributed days.** Forensic schedule analysis is exactly that labelling process — mature, commercial, standardised under AACE 29R-03 — but its outputs live in claims files and arbitration bundles.

A second, independent search reached the same conclusion against a stricter test requiring activity identity, baseline-versus-actual timing, an adjudicated cause, and a documented access route. Seven strong candidates, none qualifying. **Schedule histories have no causes, cause studies have no activities, and adjudications have no released schedules.**

Near-misses, and why each fails

Source	What it genuinely has	Why it does not qualify
E-verse Buildings.Historical.Data	70+ building projects, Primavera to CSV. Inspected `TASK.csv` carries `task_id`, `proj_id`, `wbs_id`, actual and target dates. MIT licence stated.	No cause field. Check licence against upstream provenance.
nPlan corpus	750,000+ schedules, 2 billion activity nodes.	States the corpus "consists solely of schedules." A product, not a licence.
SmartPM	Activity-level baseline vs. actual, delay drilldown listing days contributed to critical path.	Analyses *your* uploaded schedule; "contributed to critical-path delay" is arithmetic, not a reason code (§7).
ASBCA decisions, e.g. WECC No. 60949	Adjudicated causation with days apportioned — 67 days for government-caused slab and access issues, 11 for slab replacement, 56 net after mitigation.	One unstructured decision; schedule exhibits not released. A corpus to curate.
Zenodo power-plant work-package delays	Open CC BY 4.0 workbook of longitudinal work-package delays.	Schema is duration-increase and start-delay columns only. No cause field, no WBS key.
SCOPE v0.2 (Kaggle)	Structurally the closest schema: activity IDs, events, event types, cause nodes, counterfactual fields.	**Explicitly synthetic.**
CII benchmarking	Large member benchmarking warehouse.	Policy refuses raw data to external organisations.

The evidence layer is the exception — it already exists

Two independent searches were needed to assemble this; neither found everything, and the gaps differed.

Dataset	Water tag and annotation	Scale	Licence
MBDD2025	**Leakage**, boxes in Pascal VOC XML and YOLO TXT	14,471 high-resolution UAV images of building surfaces, five defect categories. 2.5 GB on Zenodo.	CC BY 4.0 — commercial OK

Dataset	Water tag and annotation	Scale	Licence
dacl10k	**Wetspot**, **Efflorescence**, **Washouts**, plus Rust and ExposedRebars. **Pixel-level polygons**, 19 classes — the only segmentation-grade water source found.	9,920 images from real bridge inspections.	CC BY-NC 4.0 — research and demo only
CUBIT-Det	**Moisture**, boxes	5,527 images up to 8000×6000. Buildings 65%. Moisture only 6% — ~330 images.	Disputed — resolve first
Wall defects (Roboflow)	**moisture**, **mold**, boxes	376 images. No data card or provenance protocol.	CC BY 4.0 stated
Water leakage (Roboflow)	**water leak**, boxes	51 source images. Prototype seed only.	CC0 declared — unproven chain

Four further GitHub-routed sources extend the water-adjacent coverage. Note that **efflorescence and corrosion staining are moisture *indicators*, not active leaks** — they evidence water migration through concrete, which is useful and is not the same label.

Source	Tag	Scale and format	Licence — verified
`pauloengcsantos/multiclass-damage-segmentation`	**Efflorescence**, (with exposed reinforcement, rust stain, scaling, crack)	Two releases, 2 and 1,132 images, 9.03 GB, YOLO format. The largest water-adjacent corpus found. Hosted on Google Drive, not GitHub.	**Code GPL-3.0; the dataset carries no explicit licence** — "research purposes" with a citation requirement. Not CC BY. Resolve before any commercial use.
CODEBRIM via `MrtnMndt/meta-learning-CODEBRIM`	**Efflorescence**, **corrosion staining**	High-resolution bridge images with bounding boxes and prebuilt splits. CVPR 2019. Data at DOI 10.5281/zenodo.2620293.	**Non-commercial and educational only**, rights reserved by FIAS and Goethe University.
`ben-z-original/sfdd`	**efflorescence**	743 concrete inspection patches with **semantic segmentation masks**.	Academic use only; redistribution prohibited.
`oakleyyy414/M3FacadeWater`	**Water seepage**, thermal-only	Claims 674 aligned UAV RGB/thermal façade pairs with YOLO boxes.	**No licence, no verified provenance, README only.** Research lead — do not cite or build on it yet.

The thermal source is worth a note despite its weakness, because it points at the one row the optical datasets cannot fill: **subsurface moisture invisible to RGB.** Thermography is a *discriminating*

test in the §5 sense — it separates active ingress from a historic stain — which places it on the evidence side of the ladder rather than the label side.

Excluded on inspection. FloodNet labels post-flood aerial inundation, not leak, damp or mould, and states no reuse licence. Facade Building Defects labels a generic `Stain` class that is colour variation, not water ingress. A twenty-image deterioration sample (`ybknb/...`) offers water-seepage and mould-spot polygons but no full corpus and unmapped class IDs — format review only.

Label the observation, never the cause

- **Observed, and directly supported:** `visible_leakage`, `visible_moisture_or_dampness`, `visible_mould`.
- **Not inferable from any of these sources:** `roof_failure`, `pipe_failure`, `facade_ingress`, `construction_defect`, `weather_damage`, and above all `schedule_delay_cause`. Establishing those needs inspection findings, repair tickets, water-test records or adjudicated cause data — which is precisely the discriminating test in §5.

  Three controls before any of this is trusted.

  Domain transfer. These are overwhelmingly exterior UAV façade images. A detector trained on them should not be assumed to work on interior drywall, ceiling stains, plumbing cabinets or insurer photographs — which is most of what a construction team photographs.

  Split leakage. Several sets ship resized and flipped augmentations. Deduplicate at source-image and structure level *before* splitting train and test, or reported accuracy measures memorisation.

  What it is licensed to do. A visual detector is a screening aid, not a forensic determination of water source, responsibility, severity or schedule impact — and saying so in the model card is what keeps §12's liability question from arriving by accident.

Proxy datasets: components and a collection protocol, not a corpus

A third search asked what exists even loosely adjacent — any operational, video or document source. Four are real and downloadable, and one of them is closer to cause data than anything else found.

Source	What it genuinely provides	Why it is not a delay corpus
Time-Motion Data Set of Construction Work (Aalto/VTT)	Manually coded MEP-worker activity segments from real site observation, with durations, waste categories and **missing-precondition categories** — materials, equipment, space, connecting work, external conditions.	A missing precondition is a *local operational condition*, not an adjudicated cause of schedule slip. No WBS ID, baseline, actual finish or delay days. Licence is inconsistent between the record page (CC BY-SA) and the rights panel (CC BY 4.0) — resolve with the depositor.

Source	What it genuinely provides	Why it is not a delay corpus
Construction/Project Management Report Examples (Kaggle)	Field-app issue tasks across several sites with project, location, description, created/target/status dates, overdue flag, package, type, and a `Cause` category.	`Cause` is an issue-task category; `Target` and `OverDue` are task-management signals, not baseline/actual schedule fields. No task-to-WBS join. CC BY-NC-SA 4.0 — not a commercial training source.
Construction Motion Data Library	6,131 processed 3D skeleton-motion samples, 4,333 construction-related, with Production / Unsafe / Awkward / Common categories. CC BY 4.0.	Controlled action-recognition data. No project, work-package, schedule, disruption or causal field.
ConSLAM	Repeated real site RGB/NIR, LiDAR, IMU and ground-truth scans, captured periodically on the same floor.	Capture time but no activity ID, planned/actual dates or cause label. Software licence is academic-use-only; data rights need confirming.

> **Do not fuse these into a root-cause model.** They share no verified `project_id`, activity key, date definition or adjudication protocol. A model that appeared to predict "cause of delay" after such a merge would be learning an artificial correspondence between a skeleton action class, a coded missing precondition, an overdue issue task and a change in point-cloud geometry. These are four different observations, not four views of one thing.

What the portfolio *is* good for is building option A's components separately — workflow-interruption analytics, constraint-text triage, action recognition, site-state sensing — and, more valuable than any of it, **designing the field-collection protocol** that would eventually produce the real joined corpus. The Aalto missing-precondition taxonomy is a well-tested starting vocabulary for that.

> **A caution for the adjudicated tier.** A second public claim opinion, *Lodge Construction v. United States*, contains genuine activity-day reasoning — and the court found the contractor's dewatering-inefficiency analysis unreliable and inflated. Adjudicated does not mean accepted. The `causal_adjudication` record in §7 must therefore store the *outcome* of the analysis, not merely that one was performed; a rejected expert allocation is a negative label, and treating it as ground truth would poison the corpus.

The combination worth building

- **OR&S 231 projects** as the baseline-vs-as-built spine, at the §6 unit.
- **E-verse Primavera exports** alongside — raw task tables, closer to the native artefact.
- **Weather joined on location and date**, giving the §4 confounder for free.
- **DOT reason codes** as the label schema, borrowed from production use.
- **AACE 29R-03 MIPs** as the adjudication protocol.
- **MBDD2025 and dacl10k for defect detection, SODA for site objects** — downloadable now, so option A starts from a trained detector.
- **ASBCA decisions** as the `adjudicated`-tier seed.
- **The labels themselves**, generated by running 29R-03 adjudication across the OR&S and E-verse data. That is the contribution.

The inter-rater experiment in §11 can run next week on data already downloadable. No acquisition, no negotiation, no partner.

Screen any vendor who claims to have this

1. Is the cause recorded at event level, or inferred by your product?
2. Who assigned it — site team, owner, expert scheduler, adjudicator, or a model?
3. Can one event link to many activities, and can an activity carry concurrent causes? What allocation rule?
4. Is the historical corpus itself licensable, or only analysis of schedules we upload?

The two open leads, now closed

Lead	What it actually is	Verdict
The 180-project corpus (Vazquez, Pozzana, Kalogridis & Ellinas, *Physica A*, 2024)	**180 real construction schedules, 108,745 delayed activities** — the largest activity-level delay corpus identified anywhere. Owned by Nodes & Links Ltd. The companion paper's availability statement: not publicly available under the terms its owners granted, but **available from the corresponding author on reasonable request**.	Real, proprietary, obtainable by asking
CMPD (*Scientific Data*, 2024)	Consolidates eleven project databases — PSPLIB, MMLIB, RG, SMCP/SMFF — which are **artificial instance libraries for benchmarking scheduling algorithms**.	Wrong object — drop it

One action worth taking before the meeting. Email the Nodes & Links corresponding author and ask for the 180-project dataset. Temper expectations twice over: the papers model delay through statistical "x-factors" — risk covariates, not adjudicated causes — so it is almost certainly a schedule-only corpus needing the labelling layer on top; and a commercial owner sharing for an academic study may not extend the same terms to a commercial project. Ask about permitted use explicitly. Even so, a yes replaces a year of substrate collection.

15. What to bring to Adrian

- **Lead with "we scale and measure RCA," not with a teardown.** The reframe in §2–§3 requires no claim that the incumbents got it wrong — only that the method cannot run often enough or compute a denominator. That is a far easier sentence to say to a team that has been working on this.
- **Concede the quantitative point up front** (§1). FTA and FMEA *do* quantify. Saying so first is what earns the narrower, correct claim about observational data.
- **Separate the two tracks** (§5). Defects are forensic and the team's instinct is right there; delay attribution is statistical and it is not.
- **Propose three concrete actions**, all startable this week: instrument the existing loop so it logs hypotheses, evidence and test outcomes; run the inter-rater measurement on downloadable data; email Nodes & Links.

- **Bring six options, not one idea** — recommend A → B with F alongside, name D as the research bet, and put the third-party reporting model (§9) on the table as F's real shape.
- **Flag insurance as an explicit scope boundary** requiring a decision rather than a drift.

Appendix A. Method and sources

Findings in §14 come from two independent searches run on 9–10 September 2026: a direct search in which each source was fetched and read, and a second pass applying a stricter four-part qualification test. Both concluded no qualifying public dataset exists. Neither search alone was complete — the first missed MBDD2025's deposit and licence, the second missed dacl10k entirely.

The RCA literature in §2, §10 and §11 was added after a review that challenged an earlier draft's claim that classical RCA "produces no quantity." That claim was wrong for FTA, FMEA and Pareto and has been corrected. Every citation below was verified against Crossref or the issuing organisation's own site on 10 September 2026. Three items raised in that review — a *Quality Progress* article on well-formed root-cause statements, and the Apollo and TapRooT methodologies — could not be verified to a citable source and appear here only as unattributed design ideas (condition-plus-action pairing, timeline before causation, a closed terminal dictionary).

RCA method and its documented limits

1. Peerally MF, Carr S, Waring J, Dixon-Woods M. "The problem with root cause analysis." *BMJ Quality & Safety* 2017;26(5):417–422. DOI 10.1136/bmjqs-2016-005511
2. Card AJ. "The problem with '5 whys'." *BMJ Quality & Safety* 2017;26(8):671–677. DOI 10.1136/bmjqs-2016-005849
3. Classen DC, Resar R, Griffin F, Federico F, et al. " 'Global Trigger Tool' Shows That Adverse Events In Hospitals May Be Ten Times Greater Than Previously Measured." *Health Affairs* 2011;30(4):581–589. DOI 10.1377/hlthaff.2011.0190
4. RCA² : Improving Root Cause Analyses and Actions to Prevent Harm. National Patient Safety Foundation, 2015; Action Hierarchy tool. https://www.ihi.org/resources/tools/rca2-improving-root-cause-analyses-and-actions-prevent-harm
5. Leveson N. STAMP, CAST and STPA; CAST Handbook and *Engineering a Safer World*. MIT Partnership for Systems Approaches to Safety and Security. https://psas.scripts.mit.edu/home/materials/
6. AACE International Recommended Practice 29R-03, Forensic Schedule Analysis. https://web.aacei.org/docs/default-source/toc/toc_29r-03.pdf

Precursor analysis and confidential reporting

7. US NRC Accident Sequence Precursor Program — CCDP and ΔCDP via SPAR models and SAPHIRE; precursor threshold 10^{-6}, significant precursor 10^{-3}. https://www.nrc.gov/about-nrc/regulatory/research/asp.html
8. NASA Aviation Safety Reporting System — voluntary, confidential, with protection from civil penalty and certificate suspension. https://asrs.arc.nasa.gov/overview/summary.html
9. CROSS — confidential reporting on structural and fire safety in the built environment, operated by Structural-Safety Ltd with IStructE, ICE and IFE across the UK, Australasia and the US. https://www.cross-safety.org/uk

Schedule substrates

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11. E-verse, Buildings.Historical.Data. https://github.com/EverseDevelopment/Buildings.Historical.Data
12. Vazquez, Pozzana, Kalogridis & Ellinas. "Activity delay patterns in project networks." *Physica A*, 2024 — 180 schedules, 108,745 delayed activities; proprietary to Nodes & Links Ltd. https://www.sciencedirect.com/science/article/abs/pii/S0378437124000785
13. Pozzana, Ellinas, Kalogridis & Sakellariou. "Spreading of performance fluctuations on real-world project networks." arXiv:2012.00480 — carries the data-availability statement. https://arxiv.org/abs/2012.00480

Taxonomy and adjudication

14. Florida DOT, Contract Change Order Reason Code Descriptions. https://www.fdot.gov/docs/default-source/construction/trnsport/business-rules/Contract_Change_Order_Reason_Code_Descriptions.pdf
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Evidence layer

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18. Flotzinger et al. "dacl10k: Benchmark for Semantic Bridge Damage Segmentation." arXiv:2309.00460 / WACV 2024 — CC BY-NC 4.0. https://arxiv.org/abs/2309.00460
19. dacl10k toolkit — splits, class list, polygon format. https://github.com/phiyodr/dacl10k-toolkit
20. MBDD2025 — 14,471 UAV images, Leakage class, 2.5 GB Zenodo deposit, CC BY 4.0. https://www.nature.com/articles/s41597-025-06318-5
21. CUBIT-Det and CUBIT-Seg, CUHK — moisture class; licence disputed. https://github.com/CUHK-USR-Group/Defect-Dataset
22. Multiclass damage segmentation — efflorescence among six classes; v2 is 41,132 images. Code GPL-3.0; dataset unlicensed, research use with citation. https://github.com/pauloengcsantos/multiclass-damage-segmentation
23. CODEBRIM (COncrete DEfect BRidge IMage dataset), CVPR 2019 — non-commercial and educational use only, rights reserved by FIAS and Goethe University. Data: DOI 10.5281/zenodo.2620293. https://github.com/MrtnMndt/meta-learning-CODEBRIM
24. S2DS — 743 concrete inspection patches with segmentation masks, efflorescence class; academic use only. https://github.com/ben-z-original/s2ds
25. M3T-facade — thermal water-seepage labels claimed; no licence or verified provenance. Cited as an unverified lead only. https://github.com/oakleyyy414/M3T-facade

Proxy sources (components, not a corpus)

26. Time-Motion Data Set of Construction Work, Aalto/VTT — coded activity segments with waste and missing-precondition categories; licence inconsistent between record page and rights panel.
27. Construction/Project Management Report Examples, Kaggle — field-app issue tasks with a `Cause` category; CC BY-NC-SA 4.0.
28. Construction Motion Data Library — 6,131 3D skeleton-motion samples; CC BY 4.0.

29. ConSLAM — repeated site RGB/NIR/LiDAR/IMU scans; software academic-use-only.
30. *Lodge Construction v. United States* — public claim opinion in which the court found the dewatering-inefficiency analysis unreliable and inflated. Cited as a caution, not a source.

Commercial holders and excluded sources

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